

ANALYSTLAB AFRICA

Data Science Internship Programme



Employee Attrition Analysis Business Understanding Report

By

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Project: ABC Manufacturing Ltd

Dataset: IBM HR Analytics – Employee Attrition & Performance

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1. Introduction

Employee attrition refers to the reduction of an organisation's workforce when employees leave the company and are not immediately replaced. Employee turnover can occur for several reasons, including employees voluntarily resigning, moving to other organisations, retiring, or leaving due to organisational decisions.

For ABC Manufacturing Ltd, understanding employee attrition is important because the loss of employees can affect workforce stability and the organisation's ability to achieve its business objectives. The purpose of this project is to explore employee data and identify patterns that may help explain why some employees leave while others remain with the organisation.

The analysis will use the IBM HR Analytics – Employee Attrition & Performance dataset. The dataset is described on Kaggle as a fictional dataset created by IBM data scientists and is specifically designed to support analysis of employee attrition and related workforce questions (Pavan Subhash, n.d.).

2. Why Employee Attrition Matters to a Business

Employee attrition can create workforce gaps and may result in the loss of organisational knowledge, skills and experience. When employees leave, organisations may also need to recruit and train replacements, creating additional demands on HR and management resources.

Workforce reporting can help organisations communicate information about the value and impact of their workforce to stakeholders (CIPD, 2025). In the context of attrition, understanding workforce patterns can therefore support HR and business leaders in identifying areas that may require further investigation.

High turnover can also place pressure on remaining employees when vacancies need to be covered while replacement staff are recruited and trained. For this reason, understanding the scale and distribution of attrition is an important starting point for workforce planning and retention analysis.

3. Role of Data Science in HR Decision-Making

People analytics involve analysing data about people to solve business problems. The CIPD notes that people analytics can support evidence-based decisions and can be used to identify groups with high attrition and inform retention initiatives (CIPD, 2025).

For this project, exploratory data analysis will examine variables such as age, department, monthly income, overtime and job role. Comparing these characteristics between employees who have stayed and employees who have left can reveal patterns that deserve further investigation.

The CIPD also recommends approaching people analytics systematically, including defining objectives, auditing available data, analysing the data, reporting findings and evaluating impact (CIPD, 2024). This project follows the early stages of that process by first defining the business problem and exploring the available employee dataset.

The analysis may also provide a foundation for future predictive modelling. Historical employee data can be used to develop models that estimate future outcomes; however, exploratory relationships should not automatically be interpreted as evidence of causation. The CIPD specifically distinguishes correlation from causation and notes that predictive analytics uses historical and current data to forecast future outcomes (CIPD, 2025).

4. Business Objective

The main business objective of this project is to understand employee attrition at ABC Manufacturing Ltd by analysing workforce characteristics and identifying variables that appear to be associated with employees leaving the organisation.

The analysis will specifically investigate:

- The overall characteristics of the workforce.
- Departments with higher levels of employee attrition.
- Whether age appears to influence attrition.
- Whether monthly income is associated with retention.
- Whether overtime is associated with employee attrition.
- Job roles experiencing higher levels of turnover.
- Variables that may be useful for future predictive modelling.

5. Expected Business Value

The findings from this analysis can provide ABC Manufacturing Ltd with an evidence-based understanding of employee attrition patterns. Rather than immediately developing a predictive machine learning model, the project first establishes an understanding of the business problem and the available employee data.

The insights generated from the analysis could help HR identify areas that deserve further investigation and support the development of employee-retention strategies. The CIPD emphasises that people analytics should connect people data with business needs and support better organisational decisions (CIPD, 2025).

Employee data must also be handled responsibly. The CIPD notes that people data can contain personal information protected by law and advises practitioners to ensure that processing is consistent with applicable legal requirements (CIPD, 2024). If predictive AI is introduced later, responsible AI practices are also relevant; NIST's AI Risk Management Framework provides guidance for managing AI risks and promoting trustworthy and responsible AI development and use (Tabassi, 2023).

6. Conclusion

Employee attrition is an important business issue because the departure of employees can affect workforce stability and create additional recruitment and training requirements. Understanding employee turnover therefore requires organisations to examine workforce data and identify patterns that may indicate areas requiring further investigation.

References

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